SOC Incident Report: Privilege Escalation Detection

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1 Introduction

This report documents a simulated privilege escalation attack detected using Splunk SIEM. The objective is to demonstrate SOC analyst workflow including attack simulation, detection, investigation, and response.

2 Lab Architecture

- Attacker Machine: Kali Linux
- Victim Machine: Ubuntu Server
- SIEM: Splunk Enterprise

3 Attack Simulation

3.1 Initial Access

The attacker gained access to the system using SSH.

```
ssh loki31@192.168.56.20
```

3.2 Privilege Escalation

After login, attacker escalated privileges using sudo.

```
sudo su
```

3.3 Command Execution

```
whoami
id
cat /etc/shadow
```

3.4 Screenshots

Figure 1: SSH Login to Victim Machine

Figure 2: Privilege Escalation using sudo

4 Detection in Splunk

4.1 Sudo Activity Detection

```
index=* source="/var/log/auth.log" "sudo"
```

4.2 Root Session Detection

```
index=* source="/var/log/auth.log" "session opened for user root"
```

4.3 Correlation Detection

```
index=* source="/var/log/auth.log"
("Accepted password" OR "sudo" OR "root")
```

4.4 Screenshots

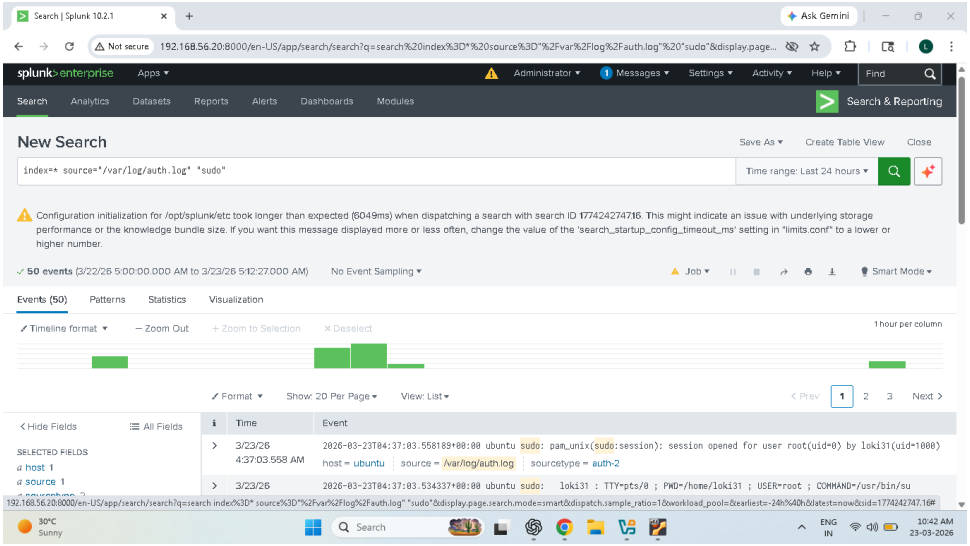

Figure 3: Privilege Escalation Detection in Splunk

4.5 Screenshots

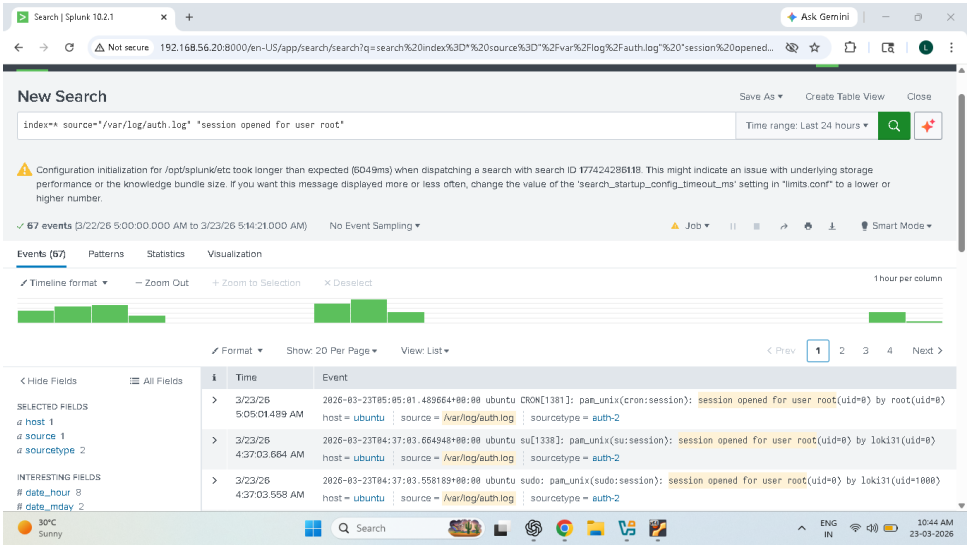

Figure 4: Privilege Escalation Detection in Splunk

5 Investigation

5.1 Timeline Analysis

```
index=* source="/var/log/auth.log"
| sort _time
```

5.2 Detailed Analysis

```
index=* "sudo" OR "root"
```

```
| table _time, user, _raw
```

5.3 Findings

- Successful SSH login detected
- Sudo command executed
- Root session established
- Sensitive file accessed (/etc/shadow)

5.4 Screenshots

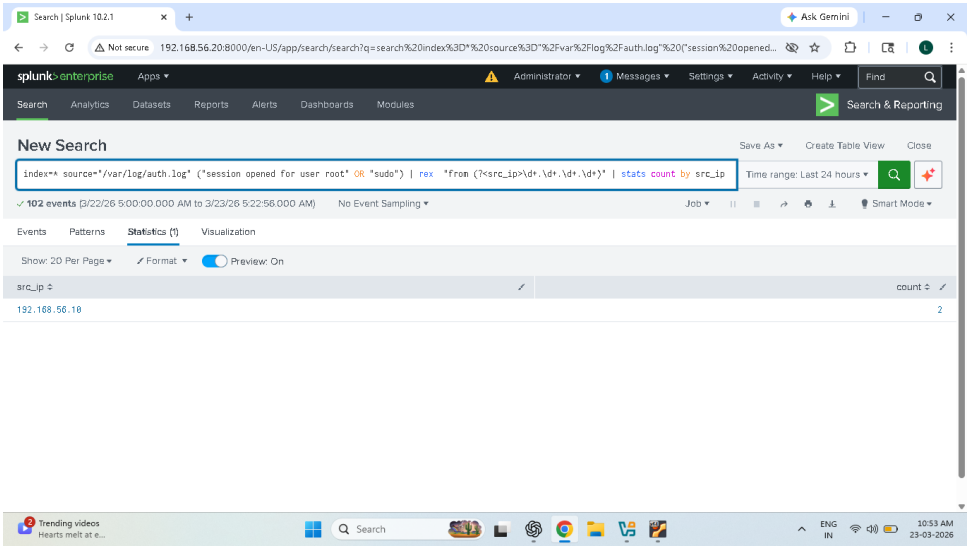

Figure 5: Timeline of Attack Activity

6 Response Actions

6.1 Account Lockdown

```
sudo passwd -l loki31
```

6.2 Restrict Privileges

```
sudo visudo
```

6.3 Firewall Blocking

```
sudo ufw deny from 192.168.56.10
```

6.4 Verification

```
sudo ufw status
```

6.5 Screenshots (Response Evidence)

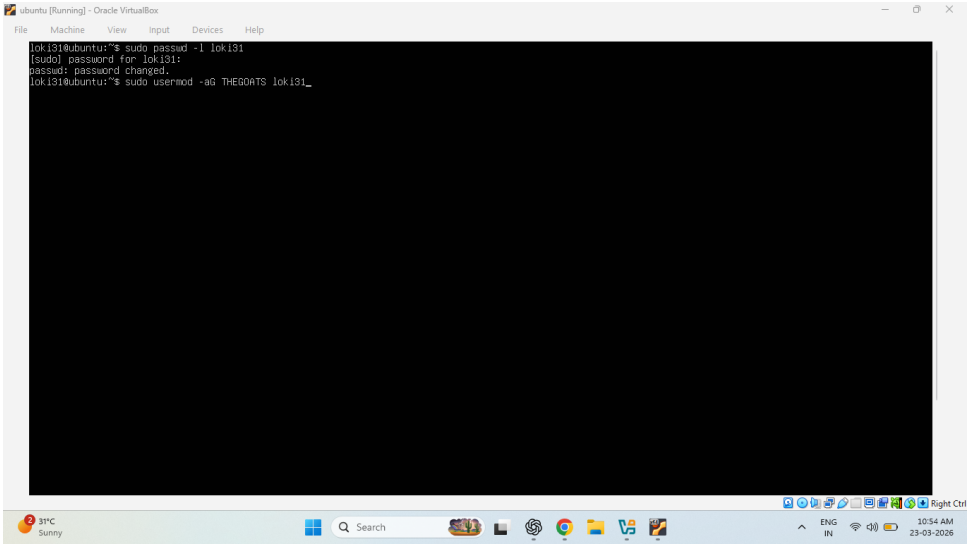

Figure 6: User Account Locked

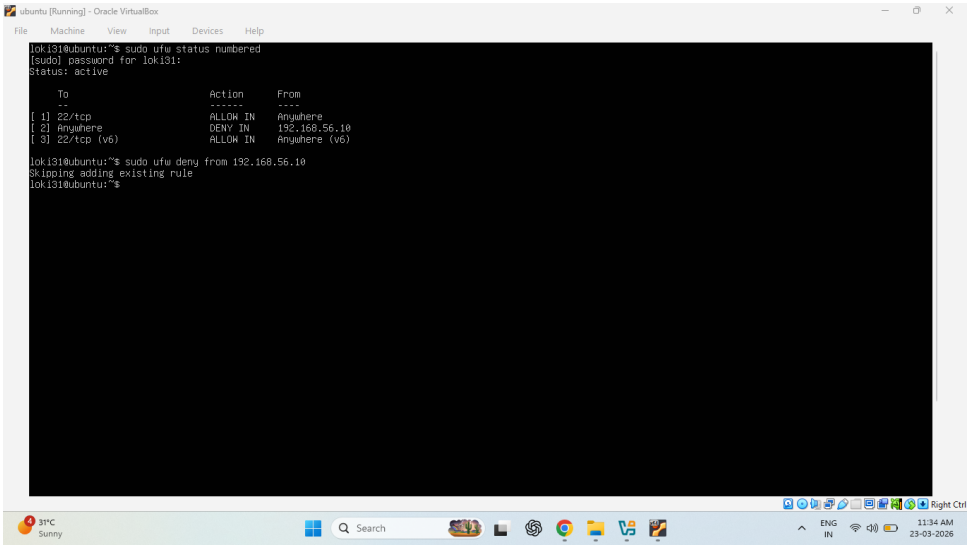

Figure 7: Blocking Attacker IP

7 Impact Assessment

- Unauthorized privilege escalation
- Full system compromise risk
- Exposure of sensitive system files

8 Recommendations

- Restrict sudo access to authorized users only

- Enable multi-factor authentication
- Monitor privileged commands continuously
- Implement least privilege principle

9 Conclusion

The incident demonstrates detection of privilege escalation using Splunk SIEM. Proper monitoring and response mechanisms were successfully implemented.